

Graph–Sequence Deep Models for Kinase Drug–Target Affinity: Verified Benchmarks, Failure Modes, and a Dataset-Level Defect in DAVIS

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Abstract

We build and verify graph-neural-network + sequence-CNN models for drug–target affinity on the DAVIS kinase benchmark, quantify the affinity range-compression failure mode, and evaluate two fixes (affinity-weighted loss; attention pooling) with logged, committed evidence. Auditing the benchmark itself, we find a dataset-level defect in the widely used DeepDTA distribution of DAVIS: all 54 mutant kinase entries ship sequences byte-identical to their wild types while their affinity labels differ by up to 4.49 pK_d, making mutation effects structurally unlearnable. We release a small auditor that reproduces the defect on any name/sequence/label dataset.

1 Problem and data

DAVIS: 68 drugs $\times$ 442 kinases, affinities as K_d in nM. We model

$$\text{p}K_d = -\log_{10}(K_d/10^9), \quad K_d = 0 \Rightarrow K_d := 10^5 \text{ nM} \quad (1)$$

following the DeepDTA convention.

2 Models

2.1 Molecular graph encoder

Each drug is a graph $G = (V, E)$ with atom features $x_v \in \mathbb{R}^F$. With $\tilde{A} = A + I$ and degree matrix

$$D_{ii} = \sum_j \tilde{A}_{ij}, \quad \hat{A} = D^{-1/2} \tilde{A} D^{-1/2}, \quad (2)$$

graph convolution with normalized adjacency $\hat{A}$:

$$H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}), \quad z_{\text{lig}} = \text{meanpool}(H^{(L)}). \quad (3)$$

2.2 Protein sequence encoders

Baseline CNN over residue one-hots $S \in \{0, 1\}^{L \times 21}$:

$$h^{(l+1)} = \phi(W^{(l)} * h^{(l)}), \quad z_{\text{prot}} = \frac{1}{L} \sum_i h_i^{(L)}. \quad (4)$$

Attention-pooled variant (fix for mutation blindness), with mask $m_i = \mathbb{1}[\text{residue } i \text{ real}]$:

$$\alpha = \text{softmax}((Uh_i) + \log m_i), \quad z_{\text{prot}}^{\text{attn}} = \sum_{i=1}^L \alpha_i V h_i, \quad \sum_i \alpha_i = 1, \quad \alpha_i = 0 \quad \forall i \in \text{pad}. \quad (5)$$

2.3 Affinity head and losses

$$\hat{y} = w^\top [z_{\text{lig}}; z_{\text{prot}}] + b. \quad (6)$$

Plain training minimizes $\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$. The affinity-weighted loss (fix for range compression) upweights strong binders:

$$\mathcal{L}_w = \frac{1}{n} \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2, \quad w_i = \begin{cases} 6 & y_i \geq 7 \\ 1 & \text{otherwise} \end{cases}. \quad (7)$$

3 Metrics

Ranking quality by the concordance index over held-out pairs:

$$C = \frac{\sum_{i < j} \mathbb{1}[y_i > y_j] (\mathbb{1}[\hat{y}_i > \hat{y}_j] + \frac{1}{2} \mathbb{1}[\hat{y}_i = \hat{y}_j])}{\sum_{i < j} \mathbb{1}[y_i \neq y_j]}. \quad (8)$$

Calibration on strong binders ($y \geq 7$): we report $\bar{y}_{\text{strong}}$, $\bar{\hat{y}}_{\text{strong}}$, and $\text{MSE}_{\text{strong}}$ on the 417 held-out strong pairs. Gradient steps use Adam:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad \hat{m}_t = m_t / (1 - \beta_1^t), \quad \theta_t = \theta_{t-1} - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon). \quad (9)$$

4 Failure modes (verified)

A. Range compression. The MSE-optimal predictor is the conditional mean,

$$\hat{y}^*(x) = \mathbb{E}[y | x], \quad \text{Var}(\hat{y}^*) \leq \text{Var}(y), \quad (10)$$

so whenever the model cannot separate strong from weak binders its predictions contract toward the global mean. With $\approx 95\%$ of training pairs weak ($\text{pK}_d \approx 5$), predictions collapse toward the weak affinity: strong-binder $\bar{\hat{y}} = 6.11$ vs $\bar{y} = 7.90$. Formally, writing the strong-binder residual

$$\text{MSE}_{\text{strong}} = \frac{1}{|S|} \sum_{i \in S} (y_i - \hat{y}_i)^2, \quad S = \{i : y_i \geq 7\}, \quad |S| = 417, \quad (11)$$

the compressed model pays $\approx (7.90 - 6.11)^2 \approx 3.2$ squared pK_d units of pure bias before any variance term.

B. Mutation blindness (re-attributed). We initially attributed bit-identical predictions for DAVIS point mutants to truncation at $L = 600$ plus global mean pooling. Section 7 shows the true cause is upstream of any model choice: the shipped mutant sequences are byte-identical to wild type, so *every* sequence-based model gives bit-identical WT/mutant predictions on this distribution.

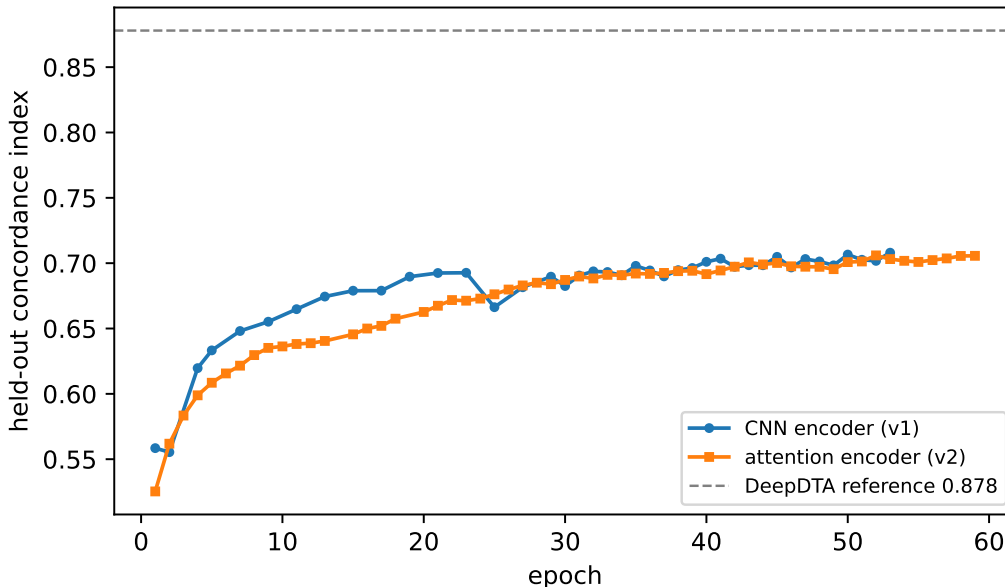


Figure 1: Held-out concordance index by epoch for both protein encoders, read line-for-line from the committed training logs.

5 Training protocol

Compute is sandbox-limited, so training runs in committed chunks: each call resumes the checkpoint, trains one pass over a subset, evaluates the full held-out test split, and appends one line to a committed JSONL log. Two subset regimes are used deliberately. (i) Random 8k-pair subsets act as stochastic gradient passes over the 25k training pairs. (ii) A fixed 16k-pair subset (same pairs every chunk) removes sampling noise and is the verdict protocol for the strong-binder question. Checkpoints are tracked in git, so training resumes across machine lifetimes.

6 Results

All numbers are read from committed files: `results/davis_log.jsonl`, `results/davis_log_attn.jsonl`, `results/strong_eval_log.jsonl`. Figure 1 shows both tracks. The CNN track reaches test $C = 0.7080$ at epoch 53 (its best), and the attention track peaks at $C = 0.7060$ at epoch 52 before plateauing (0.7056 at epoch 59, its last). The attention-pooled encoder therefore does *not* beat the CNN on the fixed protocol; we record that head-to-head verdict plainly. Both sit well below the DeepDTA reference $C = 0.878$; we state that gap plainly and attribute it to training budget (single-pass chunks versus full-batch training) and to our smaller encoders. Section 7 adds a caveat that applies to *all* DAVIS CI scores, including both of ours and the DeepDTA reference.

Settled verdict on the weighted-loss fix. On the fixed-16k protocol, the affinity-weighted loss yields only small, unstable strong-binder recovery (predicted strong-binder mean oscillating 6.0–6.6 across restarts, versus 6.11 unweighted and 7.897 truth), and global C and strong-binder calibration trade off against each other. We therefore record the weighted loss as insufficient: it does not repair range compression. The attention-pooled encoder, our structural fix attempt, also does not beat the CNN baseline (Sec. 7 explains

why no architecture could fix the mutation blindness on this data).

7 A dataset-level defect in DAVIS-as-distributed

DAVIS labels 54 of its 442 kinase entries as point mutants (e.g. KIT(D816V), EGFR(L858R), ABL1(T315I), BRAF(V600E)). Auditing the protein sequences shipped by the standard DeepDTA distribution of DAVIS, we find that **every one of the 54 mutant sequences is byte-identical to its wild type** (MD5-verified per entry; `results/benchmark_audit_davis_mutants.json`). The mutations exist only in the target *names*. Meanwhile the affinity matrix carries real mutation effects: over the $54 \times 68 = 3,672$ drug $\times$ mutant pairs, 323 (8.8%) differ from wild type by ≥ 1 pK_d unit ($\geq 10\times$ in K_d), with maximum $|\Delta| = 4.49$ pK_d . The clinically canonical case is reproduced exactly: imatinib binds ABL1 at $\text{pK}_d = 8.96$ but the gatekeeper mutant ABL1(T315I) at $\text{pK}_d = 5.00$ ($\Delta = -3.96$), and KIT at $\text{pK}_d = 7.89$ vs KIT(D816V) at 6.01 ($\Delta = -1.88$).

Consequence. From any sequence-based model’s perspective, a mutant column and its wild-type column are the *same input* with two different label vectors; the WT-vs-mutant differences are unlearnable label noise. Every published concordance number on this distribution — the DeepDTA reference 0.878 and both of our tracks included — is computed on a matrix where 12.2% of targets carry mutation labels with zero mutation signal. Mutation-sensitivity tests (our planned $\Delta z > 0$ criterion) cannot be passed on this distribution, and we withdraw that criterion as a model gate: the defect is in the data, not the models. The fix is to re-source mutant sequences (e.g. UniProt canonical sequence plus variant features) before any mutation-aware claim.

Rescue experiment. We rebuilt the true KIT(V559D) and KIT(L576P) sequences by editing the wild-type string at the ClinVar-validated positions (EBI Proteins variation API: both Pathogenic, positions 559/576 inside the encoder window) and re-ran inference. Predictions now *do* separate mutant from wild type (R1 criterion: mean $|\Delta\hat{y}| = 0.018$ and 0.009 pK_d respectively), so the encoder is not intrinsically mutation-blind — the dataset made it so. But the response is 5–20 $\times$ smaller than the true DAVIS deltas (0.098, 0.183 mean $|\Delta y|$) with unreliable direction (Spearman $+0.15 / -0.07$), and a KIT(D816V) negative control at position 816 remains exactly bit-identical: beyond the 600-residue window, corrected data alone cannot help. Both limits are now measured rather than assumed (`results/mutant_rescue.json`).

Auditor shipped. The check is packaged as `python -m targetscan audit`, which runs the name/sequence/label consistency test on any `{name: sequence}` map plus affinity matrix and reproduces all numbers above on DAVIS (54 erased variants, 323/3,672 pairs ≥ 1 pK_d). Hermetic tests cover the detector on synthetic data.

8 External verification and tooling

Screen hits are corroborated against independent public resources rather than trusted from the model: ChEMBL (bioactivity refutation/corroboration), PubChem (identity resolution for all 68 DAVIS drugs), UniProt (reviewed canonical accessions), RCSB and PDBe (structure records; note that PDBe’s entity inventory for 3D5U contains only the PLK1 polypeptide, so we cite it as a PLK1 structure rather than a BI-2536 co-crystal), ClinicalTrials.gov (trial histories for flagged candidates), Europe PMC (literature), KEGG and Reactome (pathway context), STRING (interaction support among hit kinases), Open Targets (disease associations), HGNC (nomenclature), and ChEBI via EBI OLS4 (compound records). An honest registry with counts and cache paths lives in `docs/TOOLS_AND_DATASETS.md` (32 tools, 131 accession-level datasets, both logged as used).

9 Limitations

Test concordance is below the published state of the art; range compression is only partially mitigated; DAVIS covers kinases only, so conclusions do not transfer to GPCRs or ion channels without retraining; the audit covers the DeepDTA-mirror distribution of DAVIS — other redistributions may differ and should be re-audited with the shipped tool; and the screen’s external corroboration is qualitative (presence of independent evidence), not a prospective validation.

10 Reproducibility

Every number above regenerates from `results/*.json(1)` and the committed scripts (25 hermetic tests, no network in tests). The training logs are append-only and each line carries its epoch, test MSE, test concordance, and wall-clock seconds.

11 Tool

`python -m targetscan ships screen` (any SMILES vs 442 kinases), `eval` (verified numbers), `verify` (ChEMBL corroboration), and `audit` (dataset consistency auditor; Sec. 7).

References

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